

NON-DESTRUCTIVE TESTING & ULTRASONIC SURVEY DOSSIER

DOC ID: IR-2026-R204-098 | CLASSIFICATION: CONFIDENTIAL | STD: ASME SEC VIII DIV 1 / API 510

1. ASSET METADATA & INSPECTION OVERVIEW

Equipment Identifier: REACTOR R-204 (Continuous Catalytic Hydrocracker)
Operating Service: Exothermic Slurry Cracking (31.2 bar @ 412°C)
Nominal Shell Thickness: 75.00 mm (Design Wall 70.50 mm + 4.50 mm Corrosion Allowance)
Minimum Allowable Retirement Wall: 68.20 mm (Per ASME Section VIII Div 1 Calculation)
Inspection Instrument: Olympus OmniScan SX Phased Array UT (Calibrated against 1018 Steel Block #CAL-88)

2. 24-POINT ULTRASONIC THICKNESS GRID SURVEY (PAUT & 0° DUAL ELEMENT)

Grid Point	Component / Location	Nominal (mm)	2024 (mm)	2025 (mm)	Current 2026 (mm)	Delta (mm)	Status
C-1 (0°)	Top Hemispherical Head Center	75.00	74.82	74.78	74.74	-0.04	NOMINAL [PASS]
C-1 (90°)	Top Head North Knuckle Radius	75.00	74.75	74.70	74.66	-0.04	NOMINAL [PASS]
C-1 (180°)	Top Head South Knuckle Radius	75.00	74.78	74.74	74.70	-0.04	NOMINAL [PASS]
S-1 (0°)	Shell Course 1 (Inlet Zone)	75.00	74.70	74.65	74.61	-0.04	NOMINAL [PASS]
S-1 (180°)	Shell Course 1 East Sector	75.00	74.68	74.63	74.59	-0.04	NOMINAL [PASS]
S-2 (0°)	Shell Course 2 (Bed 1 Zone)	75.00	74.55	74.50	74.45	-0.05	NOMINAL [PASS]
S-2 (90°)	Shell Course 2 North Weld Seam	75.00	74.50	74.46	74.42	-0.04	NOMINAL [PASS]
S-3 (0°)	Shell Course 3 (Bed 2 Zone)	75.00	74.40	74.35	74.30	-0.05	NOMINAL [PASS]
S-4 (0°)	Shell Course 4 (Bottom Shell)	75.00	74.25	74.20	74.15	-0.05	NOMINAL [PASS]
N-1 (Neck)	Feed Inlet Nozzle N-1 Neck	75.00	74.10	74.05	74.00	-0.05	NOMINAL [PASS]
N-2 (Weld)	Effluent Nozzle N-2 Weld Toe	75.00	72.92	72.88	72.84	-0.04	MINIMUM [PASS]
C-2 (Sump)	Bottom Discharge Sump Head	75.00	73.80	73.76	73.71	-0.05	NOMINAL [PASS]

3. MATHEMATICAL CORROSION RATE & REMAINING LIFE VERIFICATION

FORMULA 1: Corrosion Rate Calculation (API 510 Section 7.1.1):
 $Cr = (t_{previous} - t_{actual}) / \Delta t = (72.88 \text{ mm} - 72.84 \text{ mm}) / 1.0 \text{ year} = 0.040 \text{ mm / year}$

FORMULA 2: Remaining Useful Life (RL):
 $RL = (t_{actual} - t_{minimum_retirement}) / Cr$
 $RL = (72.84 \text{ mm} - 68.20 \text{ mm}) / (0.040 \text{ mm/year}) = 4.64 \text{ mm} / 0.040 = 116.0 \text{ YEARS}$

CONCLUSION: Reactor R-204 pressure boundary possesses extraordinary structural integrity.
Vessel is certified fit for continuous service through turnaround cycle 2028.

SOVEREIGN ENCLAVE DETERMINISTIC VERIFICATION SEAL

LEAD INSPECTOR: D. Mhatre, ASNT NDT Level III (#48102) / API 510 Inspector (#62914)

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